Algorithmic Trading System: RELIANCE.NS ($T+5$ Directional Forecast)

📌 Project Objective

To conceptualize, build, and evaluate a machine learning-based trading system that predicts the price direction of **Reliance Industries (RELIANCE.NS)** over a 5-day trading horizon. The system executes a Long/Short strategy based on binary classification signals.

📐 The Quantitative Pipeline

1. Data Sourcing & Feature Engineering

Raw OHLCV data was transformed into **stationary features** to ensure the model learns repeatable patterns rather than specific price levels.

- **Momentum:** Relative Strength Index (RSI) to identify overbought/oversold regimes.
- **Trend:** Distance from 20-day and 200-day Moving Averages ($Price / MA - 1$).
- **Volatility:** Bollinger Band width and distance from upper/lower bands.
- **Returns:** Daily Log Returns and Volume percentage change.

2. Machine Learning Architecture

- **Model:** XGBoost Classifier (Gradient Boosted Trees).
- **Target Variable:** Binary (1 if $Close_{T+5} > Close_T$, else 0).
- **Hyperparameter Optimization:** Conducted via a structured search to balance the bias-variance tradeoff and prevent overfitting on noise.

3. Backtesting Strategy (Long/Short)

The simulation adheres to strict institutional trading rules:

- **Execution:** Enter at **Open** price of $T+1$ following a signal at **Close** of T.
- **Holding Period:** Fixed 5 trading days.
- **Capital Allocation:** 100% of current portfolio value per trade.
- **Shorting:** Fully supported (Profit from $Price_{Entry} > Price_{Exit}$).

🛡️ Anti-Leakage Protocols (Look-Ahead Bias)

To ensure the validity of the results, the following safeguards were implemented:

1. **Chronological Split:** Training data is strictly from **2011–2020**, and testing data is **2021–2024**. No random shuffling was used.
2. **Standardization Integrity:** The StandardScaler was fitted **only** on the training set and applied to the test set to prevent future volatility from leaking into the past.
3. **Signal Lag:** Incorporated a 1-day lag between signal generation (T) and execution ($T+1$) to simulate realistic order fulfillment.

📈 Performance Metrics

Machine Learning Performance

- **F1-Score:** Measures the balance between Precision and Recall.
- **Confusion Matrix:** Evaluates directional bias (ensuring the model doesn't simply default to a "buy and hold" logic).

Financial Performance

- **Total Strategy Return:** Cumulative growth of the ₹1,000,000 initial capital.
- **Annualized Sharpe Ratio:** Measures risk-adjusted returns (Excess return per unit of volatility).

🛠️ Installation & Usage

1. **Clone the repository:**

Bash

```
git clone https://github.com/yourusername/quant-reliance-task.git
```

2. **Install Dependencies:**

Bash

```
pip install yfinance pandas numpy xgboost scikit-learn matplotlib seaborn
```

3. **Run the System:**

Bash

```
python Quant_task_2.py
```